Lasan Sanjula Perera

Electronic and Telecommunication Engineering Undergraduate

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Lasan-Perera

Experience

Electronics & Embedded Engineer *Part Time*

RYSERA Innovations *Dec 2025 – Present*

- Designed and developed the embedded control system for “Drop To Print”, a 3D printing automation project
- Designed custom PCBs and implemented firmware for microcontroller based motion and system control

Teaching Assistant [ ] *Part Time*

Mathscriber *Aug 2023 – Mar 2025*

- Tutored students in advanced mathematics topics
- Provided guidance on problem-solving techniques and exam strategies.

Education

University of Moratuwa *Aug 2024 – present*

B.Sc.Eng.(Hons.) in Electronic and Telecommunication Engineering

- **Cumulative GPA (CGPA): 4.00/4.00** [ ]
- **Dean’s List:** First, Second, and Third Semesters

Ananda College, Colombo *Jan 2009 – Jan 2023*

GCE Advanced Level (Mathematics, Physics, Chemistry, General English)

- Achieved 4A (distinction) passes and ranked **11th in the Country** (out of 40,000+ candidates) with a z-score of **+2.9037**.[ ]

Online Courses

- ROS2 (Udemy)[ ] *Jan 2026*
- Machine Learning Specialization (DeepLearning.AI, Coursera) [ ] *Mar 2025*
- Mobile Sensing and Robotics (Prof. Cyrill Stachniss, University of Freiburg) *Feb 2026*

Projects

MazeRunner V1,V2,V3 -Micromouse Robot

(High-Performance Maze-Solving Robot) [ ]

- Competition-level micromouse optimized for speed, stability, and accurate maze-solving.
- Designed the **complete hardware platform, motor drivers, PCB, and integrated** STM32F411CEU6 **with encoder motors and IMU (ICM-20602)** for precise motion control.
- Tuned the system for fast runs and sharp turns while maintaining reliable sensor accuracy.
- Project won **Champions in MazeMaster 2026, 2nd runner-up in MicroMaze 2.0 & 2nd runner-up in RoboFest 2025**.

Fully Analog Line Follower Robot (Microcontroller-Free Autonomous Robot) []

- A **fully analog line-following system** using IR sensor arrays, op-amp–based signal processing, and hardware PD control (no MCU or software).
- **Designed and implemented the complete line-sensing and control hardware**, including an 8-channel TCRT5000 IR sensor array with adjustable thresholds and weighted error calculation, analog PD control, motor driver circuitry (LM324, L293D), and full power regulation and hardware integration.

6-DOF Robotic Arm *(Ongoing Project)*

- Developing a **6-degree-of-freedom robotic arm** for precise pick-and-place and manipulation tasks.
- Utilizes **closed-loop stepper motors with high-precision encoders** to achieve accurate position control and repeatability.

Adaptive Gravity-Based Smart Infusion Pump Monitor
(Real-Time IV Drop Detection & IoT Monitoring System) []

- Implemented **ESP32 firmware** with interrupt-driven IR drop detection, real-time DPM calculation, circular buffering, and a non-blocking three-state alarm system.
- Evaluated **vision-based drop sensing** feasibility (frame rate, optics, MCU limits) and selected an optimized IR-based solution.
- Project became **Finalists in Startup Spark 2.0 & Pitch Arena 2025**.

DropToPrint - 3D Printer WiFi Automation Device
(Low-Cost WiFi Upgrade for Non-Networked 3D Printers)

- Developed a low-cost IoT solution for legacy 3D printers, adding WiFi connectivity and automation without external devices.
- Programmed firmware using **ESP-IDF** with **USB OTG (CDC-ACM VCP)** to enable the ESP32-S3 to communicate with printers.
- Designed a **web-based UI** and a **fully SMD, market-ready PCB** for compactness, seamless printer monitoring and control.

Project Phoenix Cube Satellite Project (Ongoing Project)

- Stage 1, launching a high-altitude balloon into the stratosphere.
- Batch Coordinator of the project and a member of the attitude determination and control subsystem.

Multi-Task Autonomous Robot (EN2533)
(Autonomous Robot with Advanced Navigation and Object Manipulation) [|]

- An robot capable of line following, wall following, grid solving, box manipulation, color detection, ball shooting and timed interactions.
- Designed the **complete hardware platform** and implemented **C++ firmware for autonomous navigation and task execution.**

Multi-Task Autonomous Robot (SLRC 2025)
(Autonomous Robot with Advanced Navigation and Object Manipulation) []

- An robot capable of line following, grid solving, box manipulation, color detection, water related tasks and timed interactions.
- Designed the **complete hardware platform**.

Net-Warden (Network Management System for Home and SOHO)

- Device to monitor network usage and throttle speeds exceeding allocated limits.
- Implemented **firmware, backend, and web interface** using NodeJS, MySQL, Ubuntu, HTML & CSS.
- Project became **Finalists in HackVenture 1.0**.

Selected Awards

Memorial Awards (Best Student of the Physical Science Section, Ananada College) [ ] — *Dec 2023*

- Mr. C. M. Weerarathna Memorial Award
- Mr. P. De. S. Kularathna Memorial Award
- Mr. Jayantha Mallimarachchi Memorial Award

Honorable Mention at Sri Lanka Chemistry Olympiad - Institute of Chemistry Ceylon [ ] — *Dec 2023*

Best Performing Competitor - Sapentia'21 Science Competition — *Jun 2022*

Competition Experience

Champions - Sri Lanka Robotics Challenge 2026, Robotics competition organized by University of Moratuwa, Sri Lanka — *Apr 2026*

Champions - MazeMaster 2026: A micromouse competition organized by University of Uwa Welassa, Sri Lanka — *Jan 2026*

1st runner-up - InnovMind 2026: A micromouse competition organized by SLTC Research University — *Mar 2026*

2nd runner-up - MicroMaze 2.0: A micromouse competition organized by Informatics Institute of Technology, Sri Lanka — *Aug 2025*

2nd runner-up - RoboFest 2025: A micromouse competition organized by Sri Lanka Institute of Information Technology — *Sep 2025*

Finalists - Sri Lanka Robotics Challenge 2025, Robotics competition organized by University of Moratuwa, Sri Lanka — *Apr 2025*

Finalists - HackVenture 1.0, Ideathon organized by University of Kelaniya, Sri Lanka — *Aug 2024*

Finalists - Startup Spark 2.0, Ideathon organized by IESL (Institution of Engineers Sri Lanka) — *Aug 2025*

Finalists - Pitch Arena, Ideathon organized by Faculty of Engineering, University of Ruhuna, Sri Lanka — *Dec 2025*

Skills Summary

Programming Languages: Python, MATLAB, C, C++
Frameworks & Platforms: ROS2, Webots, STM32CubeIDE, Arduino, ESP-IDF
Tools & Software: Altium Designer, AutoDesk Fusion 360, Git/GitHub
Experience & Interests: Robotics and Automation, Machine Learning, Computer Vision
Languages: Sinhala (Native), English

Leadership and Volunteer Experience

Logistics Committee Lead - IEEE Student Branch UoM [ ]	*2025/2026*
Publicity Committee Member - IEEE Student Branch UoM [ ]	*2024/2025*
Logistics Committee Member, MERCon 2024[ ] - IEEE SB UoM	*Jul 2024 - Aug 2024*
Events Committee Member, MoraForesight 2.0[ ] - IEEE SB UoM	*Mar 2024 - Aug 2024*
Logistics Committee Lead, MoraForesight 3.0[ ] - IEEE SB UoM	*Jan 2025 - Aug 2025*
Vice Chairman, MoraForesight 4.0[ ] - IEEE SB UoM	*Mar 2026 - Aug 2026*
Projects Committee Member - Old Anandian Engineers' Guild, Ananda College [ ]	*2024/2025*
Vice Captain of Senior Science Quiz Team - Ananda College	*2021/2022*

References

1. **Dr. Ranga Rodrigo**
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2. **Dr. Upeka Premaratne**
   B.Sc.Eng. (Moratuwa), M.E.Sc. (Western Ontario), PhD (Melbourne), LL.B. (OUSL), Attorney-at-Law
   Senior Lecturer - Grade 1, Department of Electronic and Telecommunication Engineering,
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